

Week 1 — Assignment 2

"Build Your Home Lab & Security Audit"

Due: Alongside the Weekend Assignment | Time estimate: 2-3 hours

What you need: Your own laptop/PC, this document, internet

Why This Assignment Exists

The first assignment was about capturing and analyzing traffic. This one is about understanding your own environment — the machine you use every day. A security professional who does not understand their own system cannot protect anyone else's.

You will:

1. Audit your own computer's network configuration
2. Check what is running and listening on your machine
3. Analyze your own network topology
4. Write a mini security assessment of your home network

Task 1 — Your Machine's Network Identity (30 min)

Step 1: Gather your network information

On Windows: Open Command Prompt and run: `ipconfig /all`

On Linux/Mac: Open Terminal and run: `ifconfig` or `ip a`

Step 2: Record the following

Item	Your value
Your computer's hostname	StudyBuddy
Your IPv4 address	192.168.56.1
Your subnet mask	255.255.255.0
Your default gateway (router IP)	blank
Your DNS server(s)	192.168.0.1
Your MAC address	E4-C7-67-74-5D-E3
Are you on Wi-Fi or Ethernet?	wifi

Step 3: Answer these questions

1. What is your network's IP range? (Hint: if you are 192.168.1.105/24, your range is 192.168.1.0 – 192.168.1.255)
Range: 192.168.56.1 to 192.168.56.254
2. How many devices could potentially be on your network? 4-5 devices
3. Is your DNS server your router or an external service? My router
4. Why does the MAC address matter? (Hint: it is a Layer 2 identifier — visible even before IP communication)
It provides permanent unique physical fingerprint

Task 2 — What Is Listening on Your Machine? (30 min)

Step 1: Check open ports

On Windows: `netstat -an`

On Linux/Mac: `ss -tlnp`

Step 2: Record what you find

Local Address	Port	State	What service do you think this is?
0.0.0.135		LISTENING	Listening to any requests that maybe sent to us
127.0.0.1.49670		ESTABLISHED	Listening for requests within the computer
127.0.0.1		ESTABLISHED	Listening for requests within the computer
			(add more rows as needed)

Step 3: Answer these questions

1. How many ports are in LISTENING state? **18 ports are in listening state**
2. Are any of these ports from the Common Ports list in your cheat sheet? **80, 443, 22**
3. Is port 3389 (RDP) open? If yes, what does that mean for security? **It is open, if not there it can provide a potential entry point for attacker**
4. Is port 23 (Telnet) open? If yes, why is that a problem? **Its not open, this causes major security risks**
5. What is the difference between 0.0.0.0:PORT and 127.0.0.1:PORT? The first means service is listening on all available network interfaces, and second one means listening on only local host

Step 4: Check running processes

On Windows: tasklist

On Linux/Mac: ps aux

List the top 10 processes you see. Identify any you do not recognize.

smss.exe	800 Services	0	340 K
csrss.exe	1244 Services	0	3,872 K
wininit.exe	1356 Services	0	2,268 K
csrss.exe	1368 Console	1	4,980 K
winlogon.exe	1460 Console	1	7,000 K
services.exe	1480 Services	0	10,808 K
Lsalso.exe	1524 Services	0	1,712 K
lsass.exe	1532 Services	0	24,260 K
svchost.exe	1656 Services	0	40,584 K
fontdrvhost.exe	1672 Services	0	2,972 K
fontdrvhost.exe	1664 Console	1	6,840 K

Task 3 — Scan Your Home Network (30 min)

Step 1: Find your network range

From Task 1, you know your IP and subnet mask. Calculate your network range.

Example: If your IP is 192.168.1.105 and subnet mask is 255.255.255.0:

- Network: 192.168.56.0/24
- Range: 192.168.56.1 to 192.168.56.254

Step 2: Ping sweep your network

On Windows (PowerShell):

```
1..254 | ForEach-Object { ping -n 1 -w 100 192.168.1.$_ | Select-String "Reply from" }
```

On Linux/Mac (Terminal):

```
for i in {1..254}; do ping -c 1 -W 1 192.168.1.$i | grep "bytes from"; done
```

Note: Replace 192.168.1 with YOUR network prefix. This scan is safe — ping only checks if a host responds.

Step 3: Record what you find

IP Address	Responds to ping?	Device type (if you know)
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192.168.56.1	Yes	Router/Gateway

Step 4: Answer these questions

1. How many devices responded to your ping sweep? Only 1
2. Did you find any devices you did not expect? I didn't get any devices
3. If an attacker got onto your Wi-Fi, how many devices could they potentially reach?
Only min

Task 4 — Your Home Network Security Assessment (30 min)

Write a mini security assessment of your own home network. Use everything you learned this week. Write in your own words. Minimum 300 words.

Structure your assessment like this:

- Network Overview — What is your network range? How many devices? What types? 192.168.56.1 to 192.168.56.254
- 1. Open Ports & Services — What ports are open? Which services are running?
 - 22- ssh
 - 80- http
 - 443- https
 - 3306- mysql
- 2. Protocol Security — What protocols does your network use? Any unencrypted protocols?
 - Ssh and https
- 3. CIA Triad Analysis — Confidentiality, Integrity, Availability of your network
 - Confidentiality: HTTP and FTP traffic readable by anyone in the same network
 - Integrity: No indication of file integrity monitoring
 - Availability: single flat network
- 4. AAA Framework Analysis — Authentication, Authorization, Accounting on your network
 - Authentication: If any service (SSH, web portal) uses only passwords with no MFA, a brute force or stolen credential gives full access
 - Authorization: Flat /24 network with no subnetting means once authenticated, a user can potentially reach all 254 addresses
 - Accounting: No mention of centralized logging
- 5. Recommendations — List 5 specific improvements you could make

#	Improvement	Why it matters	How to do it
1	Enforce https everywhere	https adds an extra layer of security to data	Install SSL certificate, redirect all port 80 traffic to 443, disable HTTP on web servers
2	Enable MFA on all logins	Stolen passwords alone won't be enough to get in	Enable two-factor authentication on SSH, VPN, and any web portals using an app like Google Authenticator
3	Segment the network into subnets	A flat /24 means one breach = access to everything	Split into subnets by role: Servers /27, Admin /28, Users /26 use firewall rules to block cross subnet traffic
4	Disable Telnet and FTP, replace with SSH and SFTP	Telnet and FTP send all data including passwords in plaintext	disable Telnet/FTP services, configure SSH for remote access and SFTP for file transfers
5	Set up centralized logging and alerts	Without logs you can't detect brute force, unauthorized access, or insider threats	Enable Windows Event Viewer alerts for Event ID 4625 (failed logins)

Task 5 — Bonus: Protocol Journal (optional, +10 bonus points)

For one full day, keep a protocol journal. Every time you use the internet, write down:

Time	What I did	Protocol used	Port	Encrypted?	What would an attacker see?
09:00	Checked email	IMAP	993	Yes	Nothing — encrypted
09:15	Visited news site	HTTPS	443	Yes	Only domain via DNS
09:30	Logged into forum	HTTP	80	No	Username + password!

Fill in at least 10 entries. This teaches you to think like a security analyst in your daily life.

How to Submit

Create a document with:

1. Task 1: Your network information table + answers
2. Task 2: Your port scan table + answers + process list
3. Task 3: Your ping sweep results + answers
4. Task 4: Your written security assessment (300+ words)
5. Task 5 (bonus): Your protocol journal (10+ entries)

Name it: Week1_HomeLab_Audit_YourName.md (or .txt or .pdf)

Grading Rubric

Criteria	Points	Your score
Task 1: Network identity recorded correctly	/15	
Task 1: Subnetting questions answered	/10	
Criteria	Points	Your score
Task 2: Port scan completed and analyzed	/15	
Task 2: Process identification	/5	
Task 3: Ping sweep completed	/10	
Task 3: Security reasoning about flat networks	/10	
Task 4: Written assessment quality (300+ words)	/20	
Task 4: CIA + AAA connections accurate	/10	
Task 4: 5 practical recommendations	/5	
Bonus: Protocol journal (10+ entries)	+10	
Total	/100 (+10 bonus)	

Tips

- Do this assignment on your HOME network, not a public Wi-Fi.
- Take screenshots of your ipconfig, netstat, and ping sweep output.
- Be honest in your assessment. If your Wi-Fi has no password, write that down.
- The best security assessments are the ones that make you uncomfortable.